1. Executive Summary

- **Project Objective:** Build an end-to-end data engineering and business intelligence infrastructure to clean, analyze, warehouse, and display cross-channel marketing campaign cohorts.
- **Architecture Stack:** Python (Pandas/SQLAlchemy), PostgreSQL relational server, SQL optimization queries, and an interactive Streamlit BI application interface.
- **Key Finding:** While the Family Shopper group drives the highest raw engagement footprint, premium revenue generation is heavily concentrated within the specialized High Spender and High Income Elite brackets.

2. Statistical Data Profile & Quality Audit

The raw marketing source contains **56,000 distinct customer transaction profiles across 28 analytical data attributes**.

Data Engineering Treatments Applied:

- **Missing Value Treatments:** The Income profile parameter had missing elements. This was resolved using median value imputation to avoid skewing downstream average financial reports.
- **Temporal Tracking:** The categorical string timestamp Dt_Customer was converted into a structured datetime format to calculate Customer_Tenure_Days against the peak collection baseline.
- **Demographic Engineering:** Extracted Age by computing the difference from Year_Birth and mapped total household constraints by summing child and teenage dependencies (Kidhome + Teenhome).

3. Normalized Relational Database Design

To optimize transaction speeds and eliminate structural redundancies, the flat source schema was decomposed into a star-like structure containing four distinct normalized entities inside the PostgreSQL demo database:

1. **customers (Core Dimensions):** Holds customer metrics, including age, education level, marriage status, and geographic origin tags.
2. **customer_spend (Financial Performance Metrics):** Warehouses explicit product categories (Wines, Fruits, Meats, Sweets, Gold) alongside cumulative value metrics.
3. **customer_activity (Traffic Metrics):** Log tracking for web visits, retail store visits, catalog order actions, and deal coupons applied.
4. **campaign_responses (Conversion Matrix):** Stores target response flags (0 or 1) indicating active marketing campaign conversions.

4. Key Performance Indicator (KPI) Analysis

By querying the relational semantic view layers, the following baseline operational highlights were uncovered:

- **Core High-Value Cohort (High Spender):** Converts at an impressive **25.71% campaign response rate**, yielding a premium average category ticket size of **$1,953.00**.
- **Traffic Inefficiency Capture (Family Shopper):** Represents the largest overall group density but exhibits a low conversion performance rate of just **8.03%**, despite logging the highest digital interaction rate (**5.8 average monthly web check-ins**).

5. Strategic Marketing Recommendations

- **Implement VIP Retention Tracks:** Allocate dedicated ad spend to target the High Spender group with exclusive early-access perks to maximize customer lifetime value (LTV).
- **Launch Targeted Family Conversions:** Since the Family Shopper cohort frequently visits the website but maintains a low checkout conversion rate, introduce targeted multi-buy coupon options, value-tier bundles, and household discount parameters to lower their barrier to purchase.

- **Optimize Regional Budget Allocation:** Scale up distribution channels and marketing campaigns within high-volume regions like Spain and Canada, while deploying specialized localized acquisition tests across lower-density markets like Mexico.